

Descriptive Analysis of Solar Data

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1 introduction

Since 2022, my parents have had solar panels installed on their roof. This document provides a descriptive analysis of the solar panel data and examines several practical questions related to electricity production, household consumption, grid usage, battery use, and seasonal variation.

A common assumption is that owning a solar panel system automatically leads to substantial financial benefits. However, the actual value of such a system depends on how much electricity is produced, how much of it is used directly, how much is stored in the battery, and how much still has to be imported from the public grid. This analysis therefore takes a data-based look at the performance of the system and evaluates how efficiently the generated solar energy is used.

The main goal is to better understand the cost-efficiency of the system. At the same time, this can also be interpreted as a measure of ecological efficiency: the more locally generated solar energy is used by the household, the less dependent the household is on externally supplied electricity.

A second motivation is to assess the current degree of grid dependence. In particular, the analysis investigates how much of the household's electricity demand can be covered by solar production and battery storage, and how much still depends on electricity imported from the public grid. This is relevant not only from a financial perspective, but also from a resilience perspective: it helps answer whether the household could maintain its current electricity use during a major disruption of the German energy infrastructure.

1.1 structure of the data

Each row represents one aggregated 3-hour measurement interval. The timestamp marks the beginning of the interval. Based on the hour of the timestamp, each row is assigned to a daytime category.

```
dat$daytime <- NA

dat[dat$hour %in% c(21, 0, 3), ]$daytime <- "night"
dat[dat$hour == 6, ]$daytime <- "morning"
dat[dat$hour == 9, ]$daytime <- "late_morning"
dat[dat$hour == 12, ]$daytime <- "midday"
dat[dat$hour == 15, ]$daytime <- "afternoon"
dat[dat$hour == 18, ]$daytime <- "evening"
```

The most important variables are:

- production [Wh]: energy generated by the solar panels.
- consumption [Wh]: energy consumed by the household.
- battery_charge [Wh]: energy used to charge the 10 kWh battery.
- battery_discharge [Wh]: energy supplied by the battery to the household.
- grid_feedin [Wh]: energy exported to the public grid.
- grid_consumption [Wh]: energy imported from the public grid.
- battery_state_of_charge [%]: current battery charge level.
- direct_consumption [Wh]: solar energy consumed directly by the household.

1.2 data cleaning

The raw dataset does not contain the same number of observations for every year-month combination. This is expected in part, because months have different numbers of days. As we already know a complete day contains eight observations. With this information we can calculate that January should contain $31 \times 8 = 248$ observations, whereas February contains either $28 \times 8 = 224$ observations or, in a leap year, $29 \times 8 = 232$ observations.

The following table shows the number of observations per year and month before cleaning:

```
table(dat$year, dat$month)
```

```
##
##      1    2    3    4    5    6    7    8    9   10   11   12
## 2023    0    0  165  240  248  240  248  248  240  248  240  248
## 2024  248  232  248  240  248  240  248  226  240  248  240  248
## 2025  248  224  248  240  248  240  248  248  240  248  240  248
## 2026  248  221  248  240  208    0    0    0    0    0    0    0
```

The table shows that some months are incomplete. March 2023 is incomplete because data collection started during that month, so it is removed from the analysis.

```
dat <- dat[
  !(dat$year == 2023 & dat$month == 3),
]
```

Other months contain only a small number of missing 3-hour intervals. For these months, the missing values are approximated using the mean from the same month, day, and hour in other years. In total, 65 out of 9256 are approximated.

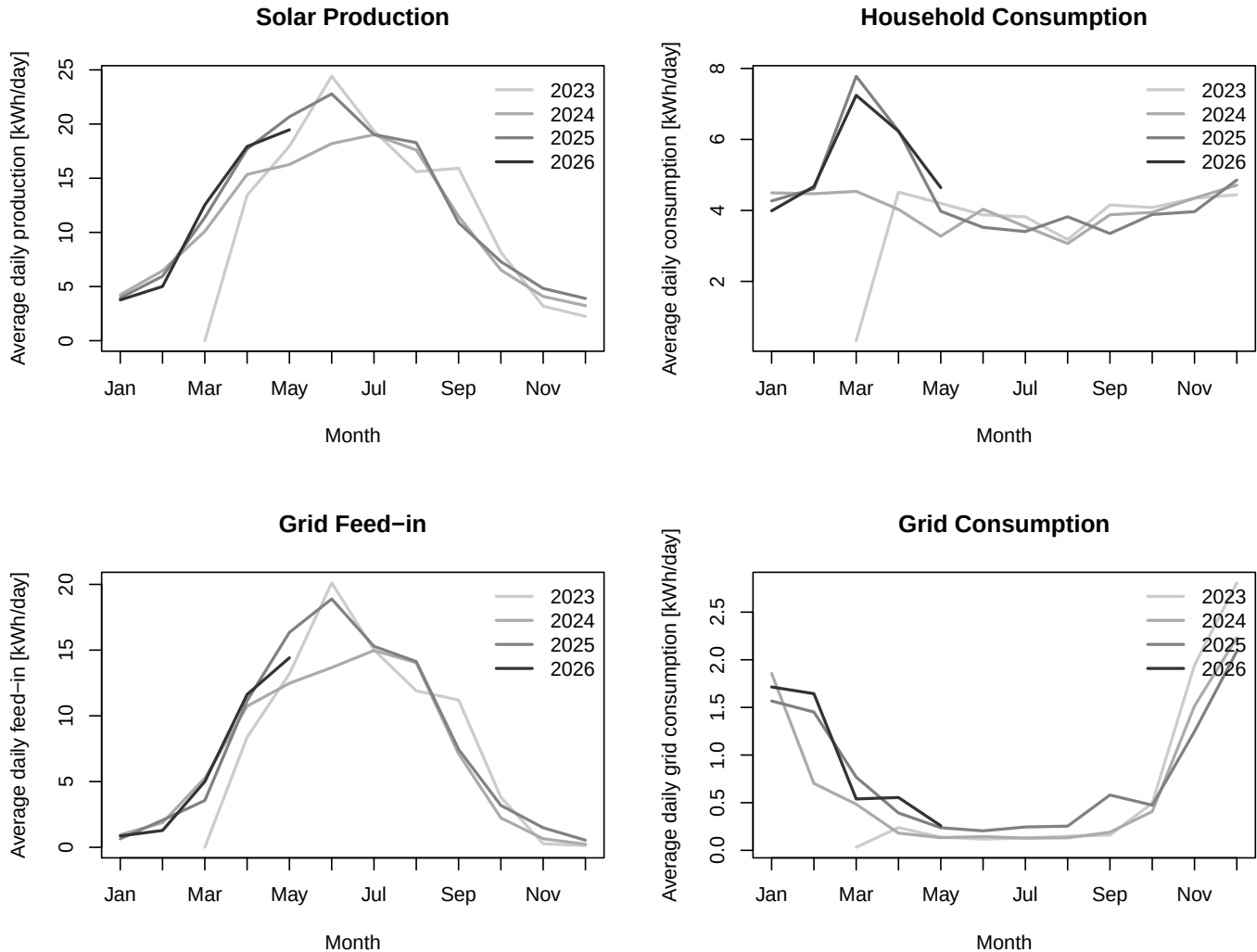
```
table(dat$year, dat$month)
```

```
##
##      1    2    3    4    5    6    7    8    9   10   11   12
## 2023    0    0    0  240  248  240  248  248  240  248  240  248
## 2024  248  232  248  240  248  240  248  248  240  248  240  248
## 2025  248  224  248  240  248  240  248  248  240  248  240  248
## 2026  248  224  248  240  248    0    0    0    0    0    0    0
```

The resulting dataset preserves the correct calendar structure of the data while removing distortions caused by incomplete months or missing 3-hour intervals. This makes seasonal comparisons of absolute production and consumption values more meaningful.

2 descriptive analysis

First I want to look at the data from a yearly perspective, and decide if you can find categories like “seasons” in the data. Because months have different numbers of days, I use average daily values for monthly comparisons instead of simple monthly sums. This makes months with 28, 30, and 31 days more comparable.



The plots show strong seasonality in the energy flows. Solar production and grid feed-in peak in the brighter months and decline sharply in winter. As a result, grid consumption is lowest during spring and summer and rises again in the darker months, when less solar energy is available.

2.1 defining solar seasons

Instead of using fixed calendar seasons such as the astronomical summer begin, I define seasons directly from the data as high-production and low-production solar months.

- **warm months:** the six months with the highest total solar production
- **cold months:** the six months with the lowest total solar production

The table ranks months by total solar production. Months with ranks 1 to 6 are classified as warm months, while the remaining months are classified as cold months. To evaluate whether high-production months also tend to have lower grid consumption, I calculated the Spearman rank correlation between the production ranking and the grid consumption ranking. The resulting correlation is -0.98.

Table 1: Monthly solar production and grid consumption ranked by production in kWh

	month	production	grid_consumption	month_name	production_rank	grid_consumption_rank	group
6	6	2724	19	June	1	12	warm
7	7	2392	21	July	2	11	warm
5	5	2323	24	May	3	9	warm
8	8	2145	22	August	4	10	warm
4	4	2013	43	April	5	7	warm
9	9	1596	39	September	6	8	warm
3	3	1417	75	March	7	5	cold
10	10	917	57	October	8	6	cold
2	2	726	157	February	9	4	cold
11	11	504	195	November	10	3	cold
1	1	498	217	January	11	2	cold
12	12	390	295	December	12	1	cold

Table 2: Total energy flows over the full observation period in MWh.

production	consumption	battery_charge	battery_discharge	grid_feedin	grid_consumption	direct_consumption
13.98	5.03	2.57	1.8	9.05	0.87	2.5

2.2 Total household performance

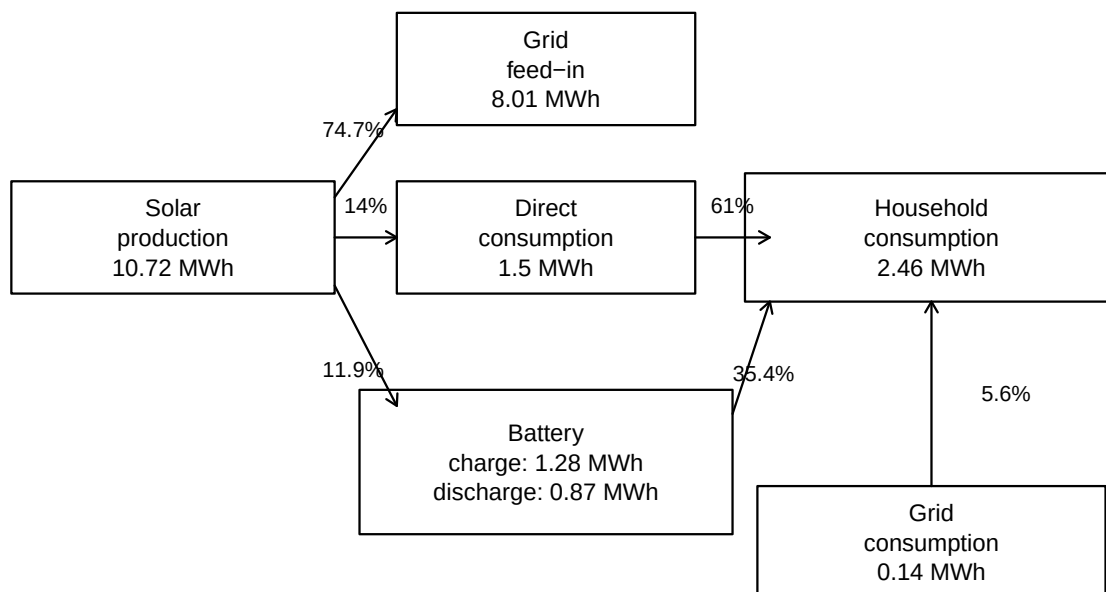
This section summarizes the total energy flows over the full observation period. Table @ref(tab:total-data) reports the absolute energy amounts in MWh. These totals describe the overall household energy balance, including solar production, household consumption, battery charging and discharging, grid feed-in, grid consumption, and the direct consumption of self-produced solar energy.

2.3 household performance by season

The total values provide an overview of the household energy balance, but they hide important seasonal differences. Solar production is expected to be higher during the warm season, while grid consumption is usually more relevant during the cold season. Therefore, the energy flows are also summarized separately for warm and cold months.

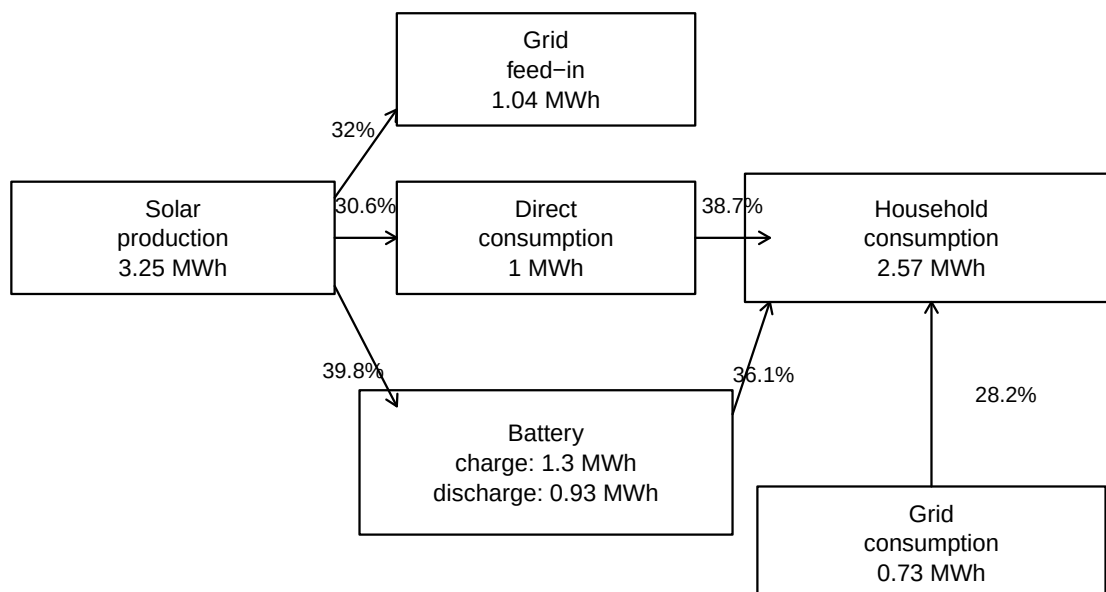
Figure @ref(fig:total-flowchart) visualizes the main energy flows between solar production, battery storage, grid exchange, and household consumption for both seasonal periods.

Total energy flows in warm months



Left-side percentages are shares of solar production; right-side percentages are shares of household consumption.

Total energy flows in cold months



Left-side percentages are shares of solar production; right-side percentages are shares of household consumption.

2.4 finance

2.5 6. What was the net energy balance of the household?

Here, the net balance is defined as total solar production minus total household consumption. A positive value means that the system produced more energy than the household consumed over the full observation period. what was the money balance?

The financial implications of grid exchange are strongly asymmetric because electricity bought from the grid is much more expensive than electricity sold to the grid. Assuming a feed-in compensation of 0.05 EUR/kWh and a grid electricity price of 0.35 EUR/kWh, one kilowatt-hour imported from the grid costs seven times as much as one kilowatt-hour exported to the grid earns.

In the warm months, grid feed-in generates about 395 EUR of revenue, while grid consumption costs only about 48 EUR. This results in a positive grid-related balance of about 347 EUR. In the cold months, however, grid feed-in generates only about 59 EUR, while grid consumption costs about 256 EUR, resulting in a negative balance of about 197 EUR.

This shows why direct consumption and battery use are financially more valuable than feeding electricity into the grid. Each kilowatt-hour used directly by the household avoids buying electricity for 0.35 EUR, whereas feeding the same kilowatt-hour into the grid would only earn 0.05 EUR. Therefore, using solar electricity locally is worth about 0.30 EUR more per kWh than exporting it.

```
total_warm$grid_feedin * 1000 * 0.05
```

```
total_cold$grid_feedin * 1000 * 0.05
```

```
total_warm$grid_consumption * 1000 * 0.35 - total_cold$grid_consumption * 1000 * 0.35
```

```
if feeding in and drawing were balanced total_warm$grid_feedin * 1000 * 0.35
```

```
net_energy_balance_mwh <- wh_to_mwh(
  sum(dat$production, na.rm = TRUE) - sum(dat$consumption, na.rm = TRUE)
)

net_grid_balance_mwh <- wh_to_mwh(
  sum(dat$grid_feedin, na.rm = TRUE) - sum(dat$grid_consumption, na.rm = TRUE)
)

data.frame(
  balance = c("Production minus consumption", "Grid export minus grid import"),
  energy_MWh = c(net_energy_balance_mwh, net_grid_balance_mwh)
)
```

```
##                               balance energy_MWh
## 1 Production minus consumption  8.943426
## 2 Grid export minus grid import  8.180250
```

Answer: The production-minus-consumption balance is about 8.9434258 MWh. The grid-export-minus-grid-import balance is about 8.1802497 MWh. This means the household produced substantially more solar energy than it consumed, but part of this surplus was exported because production and demand did not always occur at the same time.

2.6 7. What percentage of total solar production was used locally?

```
local_solar_use_wh <- sum(dat$production, na.rm = TRUE) - sum(dat$grid_feedin, na.rm = TRUE)
local_solar_share <- local_solar_use_wh / sum(dat$production, na.rm = TRUE)
local_solar_share_pct <- pct(local_solar_share)
local_solar_share_pct
```

```
## [1] 35.3
```

Answer: About 35.3% of total solar production was used locally. The remaining share was exported to the grid.

2.7 8. What percentage of household demand was covered without the grid?

This is calculated as one minus the share of consumption imported from the grid.

```
demand_covered_without_grid <- 1 - (
  sum(dat$grid_consumption, na.rm = TRUE) / sum(dat$consumption, na.rm = TRUE)
)

demand_covered_without_grid_pct <- pct(demand_covered_without_grid)
demand_covered_without_grid_pct
```

```
## [1] 82.8
```

Answer: About 82.8% of household demand was covered without importing electricity from the public grid.

3 Relationship Between Solar Production and Grid Consumption

4 Monthly Energy Balance

The monthly energy balance compares production, consumption, grid feed-in, and grid consumption within each month.

```
month_energy <- aggregate(
  cbind(production, consumption, grid_feedin, grid_consumption) ~ year + month,
  data = dat,
  FUN = sum,
  na.rm = TRUE
)

month_energy$month_name <- month.name[month_energy$month]
month_energy <- month_energy[order(month_energy$year, month_energy$month), ]

head(month_energy)
```

```
##   year month production consumption grid_feedin grid_consumption month_name
## 10 2023     4   402830    135542    251139         7168         April
## 14 2023     5   556344    130240    408882         4287          May
## 18 2023     6   732346    116166    603450         3542          June
## 21 2023     7   600427    118405    465295         4024          July
## 24 2023     8   483684     98296    369104         4586         August
## 27 2023     9   477582    124879    336202         4793       September
```

5 Cost Approximation

The financial effect of the solar panel system can be approximated using the cost of electricity imported from the grid and the compensation for electricity exported to the grid.

Assumptions:

- imported electricity costs approximately EUR 0.35 per kWh
- exported electricity earns approximately EUR 0.06 per kWh
- the data are measured in Wh, so values are divided by 1000 to convert them to kWh

```
grid_price <- 0.35
feedin_price <- 0.06

month_energy$grid_cost_eur <- month_energy$grid_consumption / 1000 * grid_price
month_energy$feedin_revenue_eur <- month_energy$grid_feedin / 1000 * feedin_price
```



```
month_energy$net_grid_cost_eur <- month_energy$grid_cost_eur - month_energy$feedin_revenue_eur

head(month_energy)
```

```
##      year month production consumption grid_feedin grid_consumption month_name
## 10 2023     4    402830     135542     251139         7168      April
## 14 2023     5    556344     130240     408882         4287       May
## 18 2023     6    732346     116166     603450         3542       June
## 21 2023     7    600427     118405     465295         4024       July
## 24 2023     8    483684      98296     369104         4586     August
## 27 2023     9    477582     124879     336202         4793   September
##      grid_cost_eur feedin_revenue_eur net_grid_cost_eur
## 10          2.50880          15.06834          -12.55954
## 14          1.50045          24.53292          -23.03247
## 18          1.23970          36.20700          -34.96730
## 21          1.40840          27.91770          -26.50930
## 24          1.60510          22.14624          -20.54114
## 27          1.67755          20.17212          -18.49457
```

This gives a rough estimate of the monthly cost after accounting for feed-in compensation. A complete financial analysis would also need installation costs, maintenance costs, battery costs, panel degradation, electricity price changes, and changes in feed-in compensation.

6 Core Questions

This chapter answers: **What kind of energy household are we looking at?**

6.1 1. How much solar energy was produced in total during the observation period?

```
total_production_mwh <- mwh(sum(dat$production, na.rm = TRUE))
cold_production_mwh <- mwh(sum(season[season$group == "cold", "production"], na.rm = TRUE))
warm_production_mwh <- mwh(sum(season[season$group == "warm", "production"], na.rm = TRUE))

data.frame(
  period = c("Total", "Cold months", "Warm months"),
  production_MWh = c(total_production_mwh, cold_production_mwh, warm_production_mwh)
)
```

```
##      period production_MWh
## 1      Total          13.977
## 2 Cold months           0.004
## 3 Warm months           0.013
```

Answer: The solar system produced about 13.977 MWh in total. Most production came from the warm, high-production months, with about 0.013 MWh, compared with about 0.004 MWh in the cold, low-production months.

6.2 2. How much electricity was consumed by the household in total?

```
total_consumption_mwh <- mwh(sum(dat$consumption, na.rm = TRUE))
total_consumption_mwh
```

```
## [1] 5.034
```

Answer: The household consumed about 5.034 MWh during the observation period.

6.3 3. How much electricity was imported from the grid in total?

```
total_grid_consumption_mwh <- mwh(sum(dat$grid_consumption, na.rm = TRUE))
cold_grid_consumption_mwh <- mwh(sum(season[season$group == "cold", "grid_consumption"], na.rm = TRUE))
warm_grid_consumption_mwh <- mwh(sum(season[season$group == "warm", "grid_consumption"], na.rm = TRUE))

data.frame(
  period = c("Total", "Cold months", "Warm months"),
  grid_import_MWh = c(total_grid_consumption_mwh, cold_grid_consumption_mwh, warm_grid_consumption_mwh)
)

##           period grid_import_MWh
## 1           Total           0.866
## 2 Cold months           0.001
## 3 Warm months           0.000
```

Answer: The household imported about 0.866 MWh from the grid. Grid imports were much higher in the cold months (0.001 MWh) than in the warm months (0 MWh).

6.4 4. How much solar energy was exported to the grid in total?

```
total_grid_feedin_mwh <- mwh(sum(dat$grid_feedin, na.rm = TRUE))
total_grid_feedin_mwh
```

```
## [1] 9.046
```

Answer: About 9.046 MWh of solar energy was exported to the public grid.

6.5 5. How much energy flowed into and out of the battery?

```
total_battery_charge_mwh <- mwh(sum(dat$battery_charge, na.rm = TRUE))
total_battery_discharge_mwh <- mwh(sum(dat$battery_discharge, na.rm = TRUE))

battery_flow <- data.frame(
  flow = c("Battery charge", "Battery discharge"),
  energy_MWh = c(total_battery_charge_mwh, total_battery_discharge_mwh)
)

battery_flow
```

```
##           flow energy_MWh
## 1 Battery charge      2.575
## 2 Battery discharge    1.800
```

Answer: About 2.575 MWh flowed into the battery and about 1.8 MWh flowed out of the battery. The difference is expected because battery storage is not perfectly efficient and because the battery may start and end the observation period at different charge levels.

6.6 6. What was the net energy balance of the household?

Here, the net balance is defined as total solar production minus total household consumption. A positive value means that the system produced more energy than the household consumed over the full observation period.

```

net_energy_balance_mwh <- mwh(
  sum(dat$production, na.rm = TRUE) - sum(dat$consumption, na.rm = TRUE)
)

net_grid_balance_mwh <- mwh(
  sum(dat$grid_feedin, na.rm = TRUE) - sum(dat$grid_consumption, na.rm = TRUE)
)

data.frame(
  balance = c("Production minus consumption", "Grid export minus grid import"),
  energy_MWh = c(net_energy_balance_mwh, net_grid_balance_mwh)
)

##                balance energy_MWh
## 1 Production minus consumption      8.943
## 2 Grid export minus grid import      8.180

```

Answer: The production-minus-consumption balance is about 8.943 MWh. The grid-export-minus-grid-import balance is about 8.18 MWh. This means the household produced substantially more solar energy than it consumed, but part of this surplus was exported because production and demand did not always occur at the same time.

7 Useful Additions

7.1 7. What percentage of total solar production was used locally?

Solar production used locally is calculated as total production minus grid feed-in. This includes direct consumption and solar energy that was stored in the battery and later used in the household.

```

local_solar_use_wh <- sum(dat$production, na.rm = TRUE) - sum(dat$grid_feedin, na.rm = TRUE)
local_solar_share <- local_solar_use_wh / sum(dat$production, na.rm = TRUE)
local_solar_share_pct <- pct(local_solar_share)
local_solar_share_pct

```

```
## [1] 35.3
```

Answer: About 35.3% of total solar production was used locally. The remaining share was exported to the grid.

7.2 8. What percentage of household demand was covered without the grid?

This is calculated as one minus the share of consumption imported from the grid.

```

demand_covered_without_grid <- 1 - (
  sum(dat$grid_consumption, na.rm = TRUE) / sum(dat$consumption, na.rm = TRUE)
)

demand_covered_without_grid_pct <- pct(demand_covered_without_grid)
demand_covered_without_grid_pct

```

```
## [1] 82.8
```

Answer: About 82.8% of household demand was covered without importing electricity from the public grid.

7.3 9. How large was the average daily production and consumption?

```

daily_energy <- aggregate(
  cbind(production, consumption, grid_feedin, grid_consumption) ~ date,
  data = dat,
  FUN = sum,
  na.rm = TRUE
)

avg_daily_production_kwh <- round(mean(wh_to_kwh(daily_energy$production), na.rm = TRUE), 2)
avg_daily_consumption_kwh <- round(mean(wh_to_kwh(daily_energy$consumption), na.rm = TRUE), 2)

summary_daily <- data.frame(
  variable = c("Production", "Consumption"),
  average_kWh_per_day = c(avg_daily_production_kwh, avg_daily_consumption_kwh)
)

summary_daily

##      variable average_kWh_per_day
## 1 Production             12.07
## 2 Consumption              4.35

```

Answer: On an average day, the system produced about 12.07 kWh and the household consumed about 4.35 kWh.

7.4 10. How variable were production and consumption from day to day?

```

sd_daily_production_kwh <- round(sd(wh_to_kwh(daily_energy$production), na.rm = TRUE), 2)
sd_daily_consumption_kwh <- round(sd(wh_to_kwh(daily_energy$consumption), na.rm = TRUE), 2)

cv_daily_production <- round(sd_daily_production_kwh / avg_daily_production_kwh, 2)
cv_daily_consumption <- round(sd_daily_consumption_kwh / avg_daily_consumption_kwh, 2)

variability_daily <- data.frame(
  variable = c("Production", "Consumption"),
  sd_kWh_per_day = c(sd_daily_production_kwh, sd_daily_consumption_kwh),
  coefficient_of_variation = c(cv_daily_production, cv_daily_consumption)
)

variability_daily

##      variable sd_kWh_per_day coefficient_of_variation
## 1 Production             8.04                  0.67
## 2 Consumption            2.08                  0.48

```

Answer: Daily production varies by about 8.04 kWh per day, while daily consumption varies by about 2.08 kWh per day. Production is expected to vary more strongly because it depends heavily on season and weather.

8 Conclusion

The data show strong seasonal variation in solar production. The system produces much more energy in high-sunlight months than in low-sunlight months. By ranking months according to total solar production, it is possible to define data-based warm and cold solar seasons.

The core results indicate that the household produces substantially more electricity than it consumes over the full observation period, but still imports some electricity from the grid. This happens because solar production and household

demand do not always occur at the same time. The battery and direct consumption help reduce grid dependence, but the public grid remains important, especially in low-production months.

how many days had a 100% solar covery? How many living without the grid at all